Binomial Distribution

ORF 309 — Based on Slides07

1. Start with a Bernoulli Random Variable

A **Bernoulli RV** is any random variable that takes only the values 0 or 1.

$$X_i = \begin{cases} 1 & \text{“success” (e.g. heads), with probability } p \\ 0 & \text{“failure” (e.g. tails), with probability } q = 1 - p \end{cases}$$

A **Bernoulli Process** is an infinite sequence $X_1, X_2, X_3, \ldots$ of independent, identically distributed Bernoulli RVs, each with the same parameter p.

2. Counting Successes: S_n

Define the random variable

$$S_n = X_1 + X_2 + \cdots + X_n,$$

which counts the **total number of successes in** n **trials**. S_n takes values in $\{0, 1, 2, \ldots, n\}$.

3. The Probability Formula: $P(S_n = k)$

Step 1. Each specific sequence with exactly k successes and $n - k$ failures has probability

$$p^k \cdot (1-p)^{n-k}.$$

For example, with $n = 8$, $k = 2$: the sequence `ssffffff` has probability $p^2(1-p)^6$.

Step 2. Every such sequence has the *same* probability (just the positions of the k successes rearranged).

Step 3. Count how many distinct arrangements exist:

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!} \qquad \text{(read: “}n\text{ choose }k\text{”)}$$

For $n = 8$, $k = 2$: $\binom{8}{2} = \frac{8\times 7}{2\times 1} = 28$ arrangements.

Putting it together:

$$\boxed{P(S_n = k) = \binom{n}{k} p^k \, (1-p)^{n-k}}$$

4. Formal Definition

A random variable X with values $\{0, 1, 2, \ldots, n\}$ has the **Binomial Distribution** if

$$P(X = k) = \binom{n}{k} p^k \, (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n,$$

where $p \in [0, 1]$ and $q = 1 - p$. We write $X \sim \text{Binomial}(n, p)$.

Sanity check (probabilities sum to 1):

$$\sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = (p + (1-p))^n = 1^n = 1.\checkmark$$

5. Expected Value: $E[X] = np$

Method 1 (long way)

Plug the formula directly into $E[X] = \sum_{k=0}^{n} k \cdot P(X = k)$ and simplify algebraically. It works, but it is tedious.

Method 2 (indicator trick — the one to use)

Write X as a sum of indicator RVs:

$$X = X_1 + X_2 + \cdots + X_n.$$

By **linearity of expectation** (no independence required):

$$E[X] = E[X_1] + E[X_2] + \cdots + E[X_n].$$

Each $E[X_i] = P(X_i = 1) = p$, so

$$\boxed{E[X] = \underbrace{p + p + \cdots + p}_{n \text{ terms}} = np.}$$

6. Application to Q3a (HW3)

Setup. You draw one coin and flip it **twice** independently. Let X = number of heads.

Conditioned on the coin type, the flips are independent and each has the same p, so $X \mid \text{coin type} \sim \text{Binomial}(2,\, p)$.

Condition	n	p	$E[X] = np$
Fair coin	2	0.50	$2 \times 0.50 = \mathbf{1}$
Canadian coin	2	0.65	$2 \times 0.65 = \mathbf{1.3}$

You do *not* need to compute $P(X = 0)$, $P(X = 1)$, $P(X = 2)$ individually. The indicator trick gives the answer immediately.